Foilborne: a guide for testers

Thank you for testing Foilborne. It records your foil sessions on your Apple Watch and shows them on your iPhone: where you went, how fast, when you were flying, and what carried the board. This is the first test build, 1.0 (50). Some of it is right, some of it is not yet, and your sessions are how it gets right. This guide says how to use it, what to try, and how to tell me what happened. Every screen it mentions is pictured in the appendix at the end, *The screens*.

What you need

- An **iPhone** on iOS 26 and an **Apple Watch** on watchOS 26. Apple rates only the **Ultra** for high-speed water sports, and its Action button is the one control that works with wet hands. Other watches from the Series 6 on record all the same.
- The **TestFlight** app from the App Store, then the invitation link from the group.
- A board. Any discipline: prone, SUP, wing, kite, parawing, pump and dock start, tow and wake, surf, and eFoil or foil assist. For the motor's data an eFoil needs a VESC with a Bluetooth module (see below); without one, everything else still records.

Install

1. Open the link on the iPhone. TestFlight installs the iPhone app; the watch app installs with it. If it does not appear on the watch, open the **Watch app on the iPhone** (Apple's), scroll to **Available Apps** and tap Install beside Foilborne.
2. Open Foilborne on the watch once and allow **Location** (While Using the App, not Allow Once), **Health** (turn on everything listed) and **Motion & Fitness**. On a board with a motor it will also ask for **Bluetooth** the first time a session starts.
3. On the iPhone, **open Foilborne once before your first session**, and again after each update: until the iPhone app has been opened, the watch has nowhere to send a session. Then **Foilborne on the iPhone › Settings › Using the watch** has the same steps as this guide, always with you.

Four places are called Settings. This guide always says which one:

- **Foilborne on the iPhone › Settings**: the gear button at the top of Foilborne's session list. Using the watch, Help improve Foilborne, My gear, Wind and weather, Disciplines, Notify when a session arrives, Diagnostics.
- **Foilborne on the watch › Settings**: the last row of Foilborne's start screen on the watch. Units, Motion, Health, Water sensor, ESC over Bluetooth, Forget ESC, Low battery alert.
- **The watch's Settings app**: Apple's, the grey gear on the watch's home screen. Action Button, and Privacy & Security › Location Services.
- **The iPhone's Settings app**: Apple's. Notifications › Foilborne, among others.

The iPhone shows one sample session, a made-up wing session in San Francisco Bay, so you can see what the analysis looks like before your first ride. Delete it whenever you like.

Set up once

All on the iPhone, in **Foilborne on the iPhone › Settings**:

- **Disciplines › Modules › Free discipline:** choose your **free discipline**. It is chosen once and cannot be changed; the others can be added. In TestFlight the other modules are free too: nothing is charged, and they do not carry over to the App Store version. If a module's button says Buy and will not tap, try again later.
- **My gear:** describe your setup, with a name, your all-up weight, whether it has a motor, and your take-off speed if you know it. The analysis uses it. "Standard" is fine to begin with.
- **Help improve Foilborne:** please turn this on. See *What to share with me* below.
- **Apple Watch Ultra:** put Foilborne on the Action button; the steps are in the next section.

The Action button (Apple Watch Ultra)

The Action button is the one control that works with wet hands and gloves, so on an Ultra it is worth the two minutes to set up. The same steps are in **Foilborne on the iPhone › Settings › Using the watch › The Action button (Ultra)**, to have on the phone's screen while you do it on the watch.

Set it up, once

1. On the watch, open **the watch's Settings app** (the grey gear on the home screen, not Foilborne's), scroll to **Action Button** and open it.
2. **Action:** choose **Workout**.
3. **App:** choose **Foilborne**. It is listed with the other workout apps.
4. **First Press:** choose the **discipline** the button should start: Wing foil, eFoil, Prone foil and so on, one line for each of the nine. Pick the one you ride most.
5. Press the crown to leave the Settings app. Nothing to set in Foilborne itself: its own start screen does not change what the button starts.

To ride another discipline with the button, change **First Press** in the same place. The place (Where) and the gear are the ones Foilborne's start screen last showed, so set those there before the first press.

If the First Press list shows the same name several times, a discipline missing, or an old name after an update: **restart the watch**. It keeps its own copy of the list and only a restart refreshes it.

Using it

- **No session running: one press** starts the First Press discipline. The watch buzzes and the Glance page comes up. For the first 10 seconds after a session has ended a press does nothing, so the summary cannot be dismissed by mistake.
- **During a session: one press** makes a **mark**. The buzz comes up to 2 seconds later, because the watch waits to see whether a second press follows. To mark twice, leave more than 2 seconds between them.
- **Two presses within 2 seconds** show **End?** on the screen and mark nothing (appendix, watch: *End?*). They need not be quick; a normal double press is fine.
- **One more press ends the session**, at least 1.6 seconds after the double press (a press sooner is ignored; wait for the buzz and the End? screen). With no press within 8 seconds the watch carries on recording; Cancel on the screen does the same.
- All of this works with Water Lock on. **Holding** the button is the watch's own: its menu, then the siren. A hold never reaches Foilborne.

The banner the watch shows as a session starts can name another discipline on watchOS 26; the session records the one you set.

eFoil and foil assist: the motor's data needs a VESC

The eFoil discipline, and the motor analysis on any board with a foil assist, read the speed controller (ESC) over Bluetooth. That takes hardware in the board:

- a **VESC** speed controller (a VESC 6, a Flipsky 75200 or 75100, a Trampa, a Makerbase and the like, running VESC firmware), and
- a **Bluetooth module on its UART port**, the kind VESC Tool on a phone uses (Flipsky's and the VESC Express among them).

Production boards from Lift, Fliteboard, Waydoo and others have their own controllers and do not expose VESC over Bluetooth, so the watch cannot read them. A board built or converted on a VESC (as my Takuma is) can.

Without the hardware, everything else still records: the track, speed, flights, heart rate, energy and motion are all from the watch, and the session is analysed like any other. Only the power side is missing: the Power and Battery pages on the watch stay at *Searching for ESC*, and the phone's power analysis is empty. Pick the eFoil discipline all the same, so the ride is read as a motor ride.

With the hardware:

1. In **Foilborne on the iPhone › Settings › My gear**, give the board a motor (Foil assist, eFoil direct drive or eFoil geared 4:1), and start the session on that gear, or on the eFoil discipline.
2. Keep **ESC over Bluetooth** on in **Foilborne on the watch › Settings** (it is on unless you turned it off).
3. Switch the board on, and **close VESC Tool** on your phone or disconnect it: the module takes one connection at a time.
4. Start the session. The Power page shows *Searching for ESC*, then *Connecting to…*, then the values. The ESC takes a few seconds to answer after power-on.

Once an ESC has answered, the watch remembers it and goes back to it in every session, and reconnects by itself after a fall (Bluetooth does not pass through water, so the values stop while the watch is under; the GPS log carries on). **Forget ESC**, in **Foilborne on the watch › Settings** under ESC over Bluetooth once an ESC is remembered, is for a new board or module. The battery's charge needs the battery set up in the ESC with VESC Tool (cells in series and capacity); without it the Speed page says *Charge unknown* and there is no low battery alert. The alert's first level (50, 40, 30 or 20 %; 30 unless you change it) is in **Foilborne on the watch › Settings › Low battery alert**, shown once you ride with a motor.

Recording a session

1. On the watch, open Foilborne, choose the **discipline**, **Where** (flat water, waves or downwind) and **Gear**, and tap **Start**. On an Ultra, one press of the Action button starts the discipline you set for it (above).
2. Ride. Turn the crown or swipe between the pages: **Glance** (your speed as large as the screen allows; with a motor, the pack's charge under it), **Battery** and **Power** with a motor, **Speed**, **Session** and **Controls**. The appendix shows each page and what it is for.

3. **Mark** moments you remember, a good wave, a fall, a jibe that worked: one press of the Action button, the flag on the Speed page, or Mark on the Controls page. The marks show on the map and the timeline afterwards, so you can tell me what happened at each.
4. **End:** on the Controls page hold **End** for a second (with Water Lock on, hold the crown first), or with the Action button as above. The summary comes up; Done goes back to the start screen.
5. The session comes to the iPhone by itself. Keep the phone nearby; a notification says when it has arrived, and tapping it opens the session. If it has not come within a few minutes, see *If a session does not reach the iPhone* below.
6. **Name it.** On the iPhone, the sliders button at the top of the session opens its settings: give it a **title** ("Main Beach, first wing session"), and correct the discipline, where you rode, the gear or the wind direction if the watch had them wrong (appendix, iPhone: *A session's settings*). A title makes a session easy to point at when you report something.

Please do not install a new build while the watch is recording: it stops the session.

If a session does not reach the iPhone

1. **Check it has ended.** The summary comes up after End; until then the session is still recording. On the watch, Foilborne's start screen › **On the watch** says *recording* beside it.
2. **Open Foilborne on the iPhone**, with the watch nearby and Bluetooth on, and **pull the session list down**: that asks the watch for the sessions it holds. If the iPhone app has not been opened since it was installed or updated, this is usually all it needed.
3. **Look at On the watch** (Foilborne's start screen on the watch, under Start). Each session shows *waiting for the iPhone*, *sending to the iPhone* or *on the iPhone*. If it stays on *waiting for the iPhone*, tap **Send again**. *Large files wait for the iPhone app or its charger* is normal: the session is already on the phone, and its large motion files follow while the iPhone app is open or the phone is charging.
4. **Still not there:** send me the diagnostics (*A problem with the app itself*, below). They show what the watch and the phone did with the session.

Every ride goes into Apple Health

A session is saved as a **workout in Apple Health**, so it counts in your rings and shows in the Fitness app like any other workout: the route, your heart rate through the ride, active and resting energy, the water temperature, and each flight as a segment. eFoil sessions go in as water sports, SUP foil as paddle sports, and every other discipline as surfing sports. That is why the watch asks for Health when you first open it; say yes to everything listed, or the workout cannot be saved.

Two things to know:

- A session in a discipline you do not have the module for, and have not chosen as your free one, is **not** added to Health, even if you buy the module later. In TestFlight the modules are free, so take them, and it does not arise.
- The line under a session's tiles on the iPhone says what happened: *Saved to Apple Health as a workout*, or why not (appendix, iPhone: *The tiles and the Apple Health line*). If a session is missing from Fitness, that line is the first thing to check, and worth reporting.

To keep heart rate or energy out of Health and the log, turn them off in **Foilborne on the watch › Settings › Health**.

What to test

Everything, but these most of all:

1. **Does the watch keep recording for a whole session?** Note the time if it stops, loses GPS (the speed turns yellow with WEAK GPS) or says LOCATION OFF (appendix, watch).
2. **Is the ride read right?** Open the session on the iPhone: the timeline shows when you were flying, gliding, pumping or down. Compare it with what you remember, your marks help here. Flights, waves, tacks and jibes, falls: are they where they happened, on the timeline and the map, and are any missing or invented?
3. **Speed and distance** against anything else you track with.
4. **Weather and tide:** the wind and the tide at the top of a session, and the tide page (the waves button at the top of the session list) against what you saw.
5. **With an eFoil or a foil assist on a VESC:** the ESC's values on the watch's Battery and Power pages, the low battery alert, and the power analysis on the phone. Without a VESC, that the ride is still read right from the watch alone.
6. **The Action button:** did a double press ever end a session by mistake, or a mark get missed?
7. **Battery:** how much watch battery a session took.
8. **Apple Health:** does every session appear in Fitness as a workout, with its route, heart rate and energy?
9. **Anything that is confusing or wrong** in the words, the pages or the guide.

How to give feedback

Say which session, by its title or its date and time in the list, and when in the session, by the clock on the timeline or by a mark. Then:

- **A quick note or a question:** the WhatsApp group, or email foilborne.app@gmail.com.
- **A screenshot:** take one on the iPhone; TestFlight offers to send it with a note, straight to me.

Something happened during a ride

A fall the app missed, a flight it invented, a wave read as a tack, the speed stuck: the files of that session show me what the app saw.

1. **On the water, make a mark** when it happens: one press of the Action button, or Mark on the Controls page. The mark shows on the map and the timeline afterwards, so neither of us has to guess the time.
2. On the iPhone, open the session and tap the **share button** (the square with the arrow, top right).
3. Choose **Share files** (appendix, iPhone: *The share menu*). It makes one Foilborne file of the session's files (what is in it is listed below), encrypted on your iPhone so that only I can open it: your rides stay private on the way.
4. Send it by **Mail** to foilborne.app@gmail.com, or by AirDrop or WhatsApp, with a line on what happened and at which mark.

A problem with the app itself

The watch stopped recording, a session never reached the phone, a crash, a page that would not open, a permission that kept asking: the diagnostics log says what both devices were doing.

1. On the iPhone, **Foilborne on the iPhone › Settings › Diagnostics** (appendix, iPhone: *Diagnostics*).
2. Tap **Ask the watch for its log** and give it a moment; the watch's log comes over to the phone. The watch must be on and nearby, with Foilborne open on it: the button shows only while the watch is in reach.
3. Tap **Share diagnostics**. It makes one Foilborne file of both devices' logs and the phone's crash reports (listed below), encrypted the same way.
4. Send it to foilborne.app@gmail.com with what happened, when, and what you expected. If a session is involved, send its files as well (above).

What is in the two Foilborne files

So you know what you are sending. Your iPhone encrypts each Foilborne file to my key before it leaves the phone, so only I can open it, whether it goes by Mail, AirDrop or WhatsApp; there is no password to remember. Inside are:

A session's files (Share files) are the ride itself:

- `log.csv`: a row a second or faster with the time, your **GPS position**, speed, course, height, heart rate and energy, and on a board with a motor the ESC's readings (volts, amps, rpm, duty, temperatures, faults).
- `motion.ftmo`: the watch's motion sensors through the session (acceleration, rotation, the compass, air pressure), at a higher rate with Help improve Foilborne on, which also adds `raw-000.ftra` and on: the raw accelerometer.
- `events.jsonl`: what happened during the session: the discipline, place and gear you chose, the workout's states, the ESC link's connections and drops, water in and out, your **marks**, the watch's battery and temperature each minute, GPS quality, errors.
- `manifest.json`: the session's date, duration, the watch model and app build, and the headline numbers.
- `health.json`: whether the workout was saved to Apple Health.

The diagnostics (Share diagnostics) are the app's own notes, not the ride:

- `diagnostics.jsonl` (the iPhone) and `watch-diagnostics.jsonl` (the watch, after *Ask the watch for its log*): one line per event, newest few days. At each launch, the app build, the system version, the device model, free storage and low-power mode. Then: sessions starting and ending, the ESC link's states and the Bluetooth devices it skipped, every file's transfer and arrival, notifications posted, purchases and restores (which modules, never payment details), each analysis and how long it took, the weather and tide lookups with their results (the wind figures, the tide station's name), and every error with its reason.
- `metrickit/`: Apple's reports on crashes, hangs and heavy processor or disk use, as JSON.

The diagnostics name your sessions by their file names (dates and times), and a tide station you chose or the wind at a session say roughly where you ride. They hold no GPS track, no heart rate and nothing from Apple Health, no names and no payment details. Both files come to me only, and only I can open them; I keep them under your contributor ID, use them to fix what you reported, and delete them when you ask.

What to share with me

Turn on **Foilborne on the iPhone › Settings › Help improve Foilborne**; it asks you to agree first (appendix, iPhone). With it on, the watch records more detail on every session: faster motion sensing, the compass, air pressure, every GPS fix. It costs about a fifth more battery during the session, which in practice means little: a 2-hour session on a first-generation Ultra with it on still leaves the watch running the rest of the day and into the next. It also uses about 100 MB an hour on the watch, freed as soon as the session has moved to the phone. Then, now and again:

- **Send sessions to the developer**, which appears under Help improve Foilborne once it is on. It puts every session not yet sent in one Foilborne file for the share sheet, encrypted on your iPhone so that only I can open it; Mail to foilborne.app@gmail.com, AirDrop or WhatsApp all work. **Send the last sessions again** repeats the last one if it did not go through.
- With each batch, a few words per session help more than anything the sensors record: what the conditions were, what you were riding, what went well and what did not, and what the app got wrong.

What you send carries where you rode and an anonymous contributor ID, not your name, and travels encrypted, so only I can read it. It is used only to make the analysis right for every rider, is kept by that ID, is never passed on, and is deleted if you ask. You can turn it off at any time.

The voucher

Sharing takes your time, in the minutes to send and the lines to write, so it is paid for:

- **Five sessions** sent with Help improve Foilborne on, each with a line on what happened, and I send you a code for **any one module** of your choice.
- **Twenty sessions**, and a code for **all the modules**.

The codes are App Store offer codes, redeemed in **Foilborne on the iPhone › Settings › Disciplines › Modules › Redeem a code**, and they carry over to the App Store version, so what you earn in the test stays yours. Sessions from any discipline count, and ones where the app got it wrong count most. I will let you know in the group when you cross each line; if I lose count, remind me with your contributor ID, shown under Help improve Foilborne in Foilborne on the iPhone › Settings.

Thank you

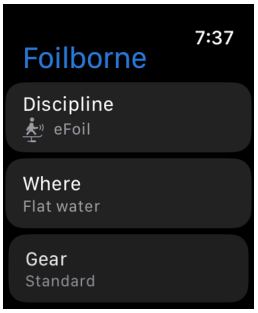
Every session you send teaches the app something no sensor can learn alone: what a real wave, a real tack or a real dock start looks like from the wrist. The voucher above is the least of it: thank you for riding with it on and telling me what happened.

Jason
foilborne.app@gmail.com

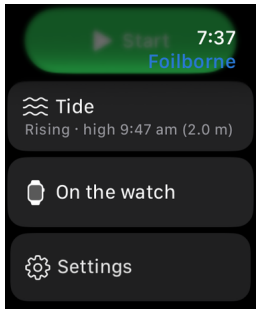
Appendix: the screens

The pictures come from the simulator, on made-up sessions; yours show your own rides.

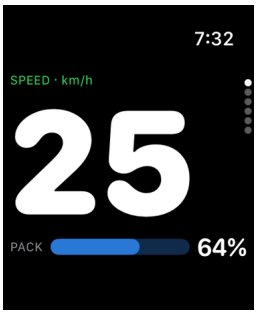
On the watch



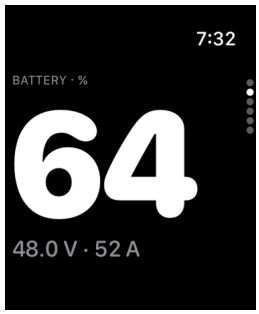
The start screen. Foilborne as it opens: the discipline, Where (flat water, waves or downwind) and the gear, then Start. Set these before you go in; the Action button starts with the Where and gear shown here.



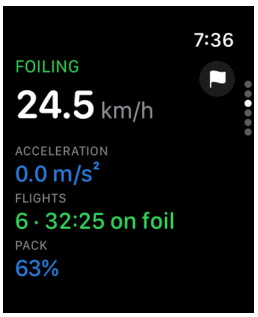
Under Start. Tide, with its gist on the row; **On the watch**, the sessions the watch holds; and **Settings**, which is *Foilborne on the watch › Settings*.



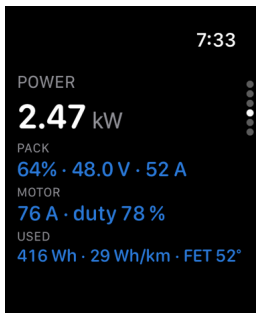
Glance. Every session opens here: your speed, as large as the screen allows. With a motor and the ESC, the pack's charge as a bar under it. The caption turns green while the board flies. The page to look at on the water.



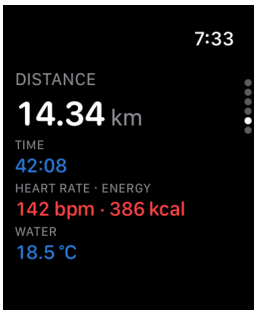
Battery (with a motor). The pack's charge in large figures, the volts and amps under it. Before the charge is known, the volts take the large figure. Look here when deciding to turn back.



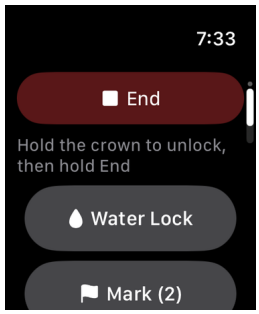
Speed. FOILING in green while the board flies (IN THE WATER when the watch is under, the pumping rate while you pump), the speed, the acceleration and load in g, your flights and time on the foil, and with a motor the pack. The flag at the top marks the moment.



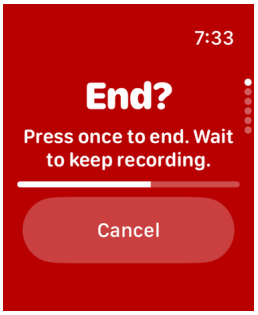
Power (with a motor). The battery's power in kW; the pack's charge, volts and amps; the motor's amps and the ESC's duty; the Wh used, Wh per km and the ESC's temperature; a fault in red if the ESC reports one. Until the ESC answers: *Searching for ESC*.



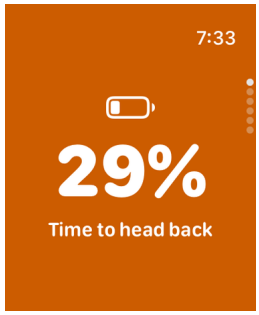
Session. The distance, the time, your heart rate and energy, and the water's temperature (Ultra and Series 10). For a look between runs.



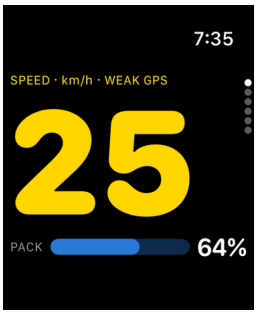
Controls. End (hold it for a second), how to end on this watch, **Water Lock**, **Mark** with the count so far, and what is being recorded. Turning Water Lock off (hold the crown) brings you here.



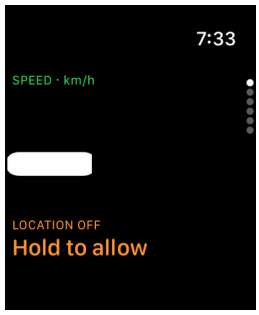
End? After a double press of the Action button. One more press ends the session; with no press, the bar runs out in 8 seconds and recording carries on. Cancel takes it back when the screen is unlocked.



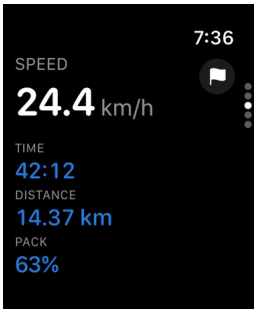
The low battery alert (with a motor). Three buzzes and the charge left on amber, at your first level (30 % unless you change it), then 20 % and 10 %, each once a session. A tap takes it away; it also goes by itself.



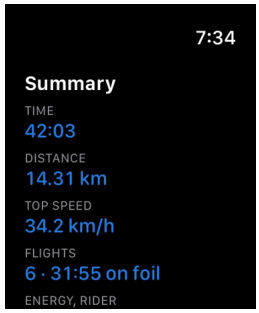
WEAK GPS. The speed in yellow: the watch is less sure of its place than usual. The speed is still good; the position may be a few metres out. A dash means no fix for 2.5 seconds, usually the watch under water.



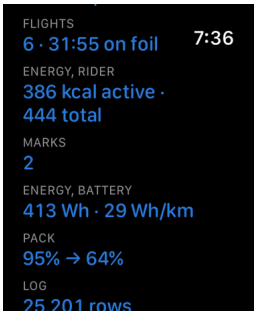
LOCATION OFF. The watch took the location away mid-session: no track or speed until it is back. Hold *Hold to allow* for a second. If it says *Allow in Settings*, fix it in *the watch's Settings app › Privacy & Security › Location Services › Foilborne*: While Using the App, Precise on.



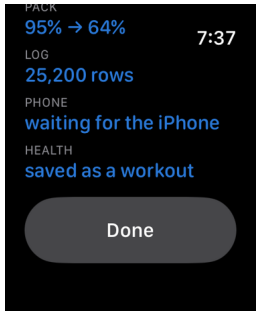
Without the module. A discipline whose module you do not have: the Speed page shows the time and distance, and there is no Session page. Everything is still recorded.



The summary, after the end: the time, distance, top speed, flights and time on the foil. Scroll for the rest (next two).



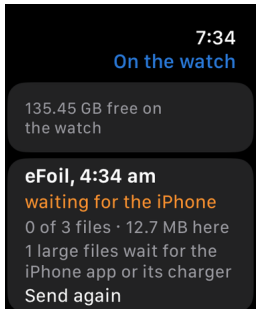
The summary, further down: your energy, the marks, and with a motor the battery's Wh and Wh/km and the pack's charge from start to end.



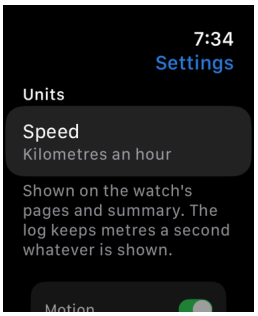
The summary's end: the log's rows, whether the session has reached the iPhone, and Health: *saved as a workout*. Done goes back to the start screen.



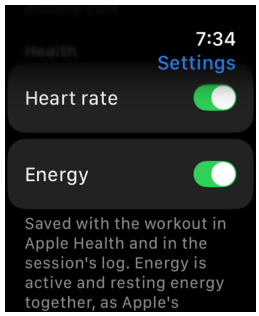
Tide (under Start). The nearest tide gauge, the height now and the next high or low, the curve from six hours ago to eighteen ahead, and the current where a gauge is near. Worked out on the watch. Not for navigation.



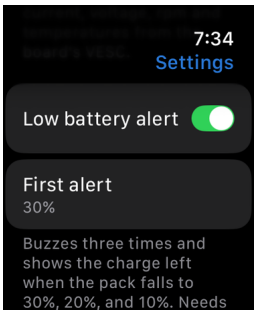
On the watch. The space free and every session the watch holds, with its state (*waiting for the iPhone*, *sending*, *on the iPhone*) and its files. **Send again** resends one. Look here when a session has not reached the phone.



Foilborne on the watch › Settings. At the top, **Units**: the speed in km/h, knots or mph. Then **Motion** and its rate. A change counts from the next session.

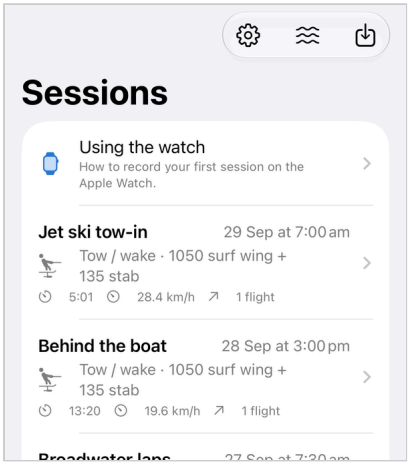


Settings, Health: Heart rate and **Energy**, saved with the workout and in the log. Further down: **Water sensor**, **ESC over Bluetooth** and, once an ESC is remembered, **Forget ESC**.

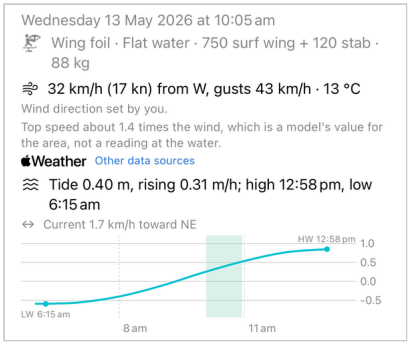


Settings, at the end: the **Low battery alert** and its **First alert** level, shown once you ride with a motor and the ESC on.

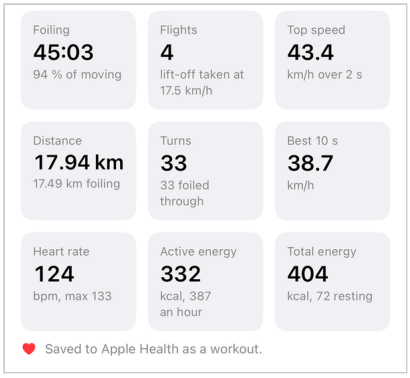
On the iPhone



The session list. At the top, the gear opens *Foilborne on the iPhone › Settings*, the waves the tide page, and the arrow imports a log. **Using the watch** heads the list until your first session from the watch is in. Each row: title, date, discipline and gear, time, top speed, flights.



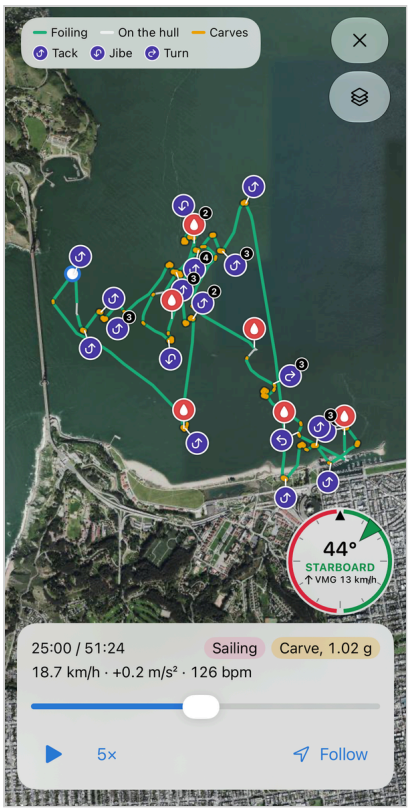
The top of a session. The date, discipline, where and gear; the wind from Apple Weather (a model's value for the area, the direction more trustworthy than the speed); and the session's tide. At the very top, the sliders button opens *A session's settings* and the square with the arrow the *share menu*.



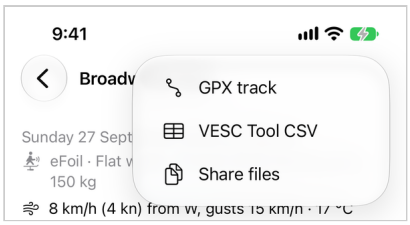
The tiles and the Apple Health line. Time foiling, flights and the take-off speed used, top speed, distance, turns, best 10 seconds, heart rate and energy. Under them, the Apple Health line: *Saved to Apple Health as a workout*, or why not. If a session is missing from Fitness, check this line first.



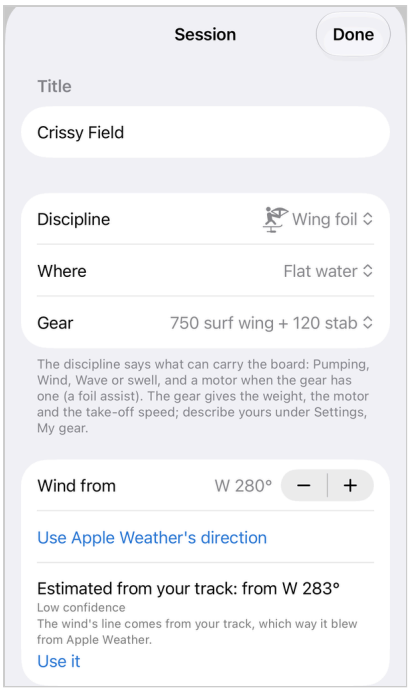
The ride: the timeline. A strip with a lane for each thing that carried the board (pumping, sailing, surfing, the motor), carves, your marks and weak GPS; then the speed, the heart rate and the acceleration with touches marked. The green bands are flights. Drag across a chart to read a moment. Compare it with what you remember.



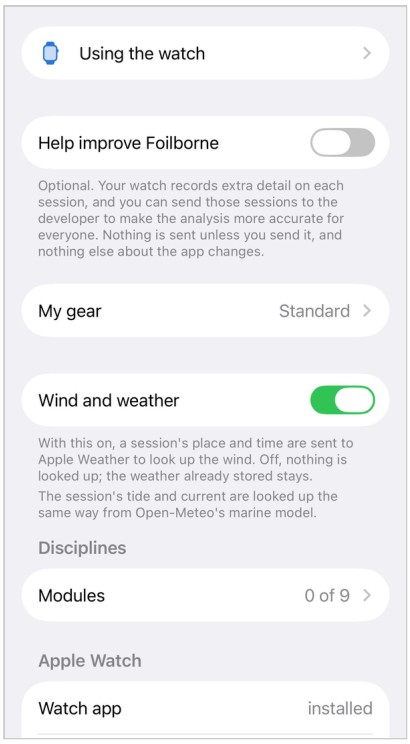
The map and its marks. Tap the session's map for this. The track green where the board flew; tacks, jibes and turns as purple badges, touches as red drops, carves in yellow, and your marks as black flags (a number where several join). The wind dial shows the wind's angle off the bow; the panel below replays the session. Tap a mark to move the replay there.



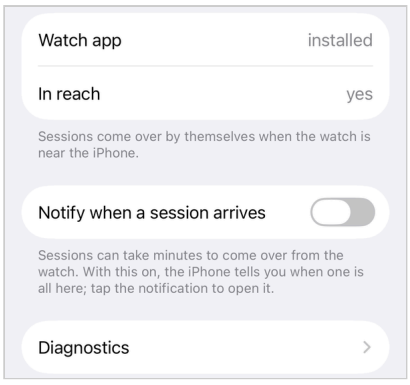
The share menu (the square with the arrow, top right of a session). **GPX track**, for Strava, Garmin Connect or a map app; **VESC Tool CSV**, only on a session with the ESC's log; **Share files**, the session's own files as one Foilborne file that only I can open (encrypted on the iPhone): the one to send me when something went wrong.



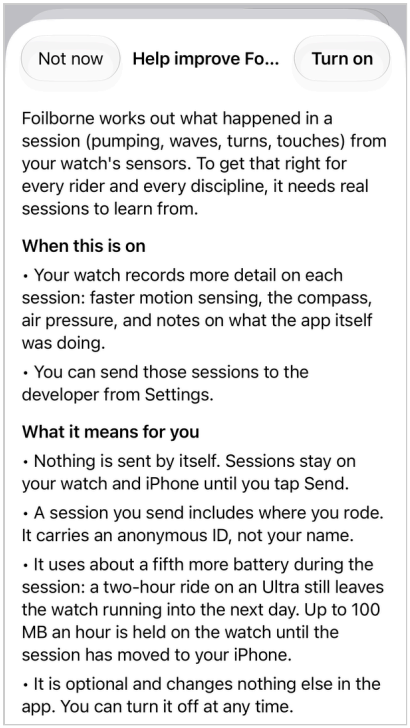
A session's settings (the sliders button at the top of a session). The **title**; the **discipline**, **where** it was ridden and the **gear**, to correct what the watch had; and **Wind from**, your own wind direction if Apple Weather's was wrong on the water, with the direction worked out from your track and **Use it**. Done saves; the session is worked out again.



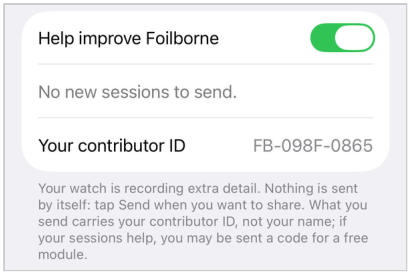
Foilborne on the iPhone › Settings. Using the watch; **Help improve Foilborne** (with it on, **Send sessions to the developer** and your contributor ID appear under it); **My gear**; **Wind and weather**; **Disciplines › Modules**; **Apple Watch**, whether the watch app is installed and in reach. Further down: **Notify when a session arrives**, **Diagnostics**, and the version.



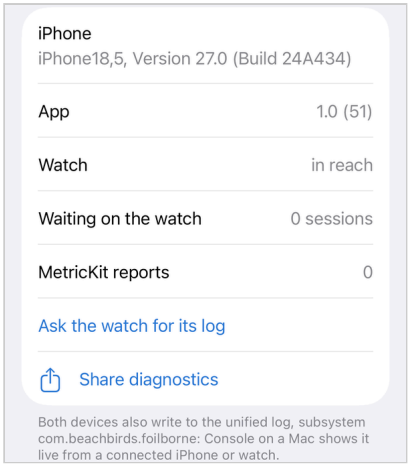
Settings, further down. Apple Watch: whether the watch app is installed and in reach. **Notify when a session arrives**: on by default (off here, in a simulator that has not allowed notifications); if the iPhone has refused them, turning it on takes you to *the iPhone's Settings app › Notifications › Foilborne*. **Diagnostics**, for reporting a problem.



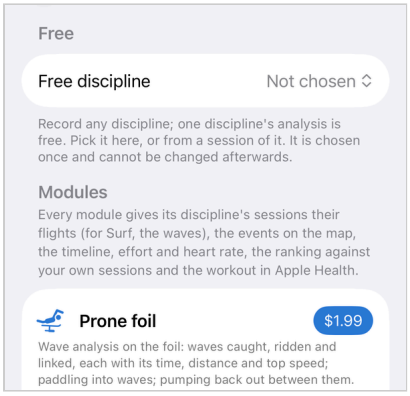
Help improve Foilborne asks you to agree first: what the watch records with it on, and what it means for you. **Turn on** at the top right.



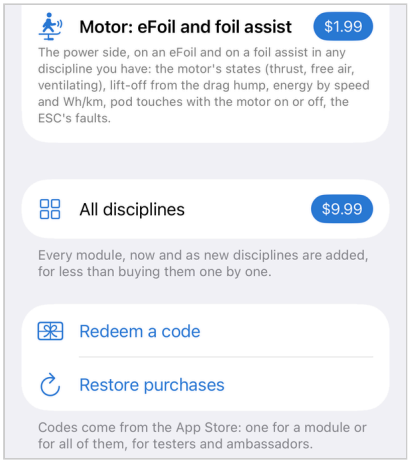
With it on: **Send sessions to the developer** under the switch once there are sessions recorded with it on (*No new sessions to send* until then), and **Your contributor ID**, the one to quote for the voucher.



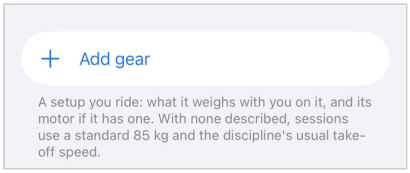
Diagnostics (Settings › Diagnostics). The devices and the build, whether the watch is in reach, what is still waiting on it. **Ask the watch for its log** shows while the watch is in reach; **Share diagnostics** makes the Foilborne file, encrypted so that only I can open it. Under them, both devices' recent events.



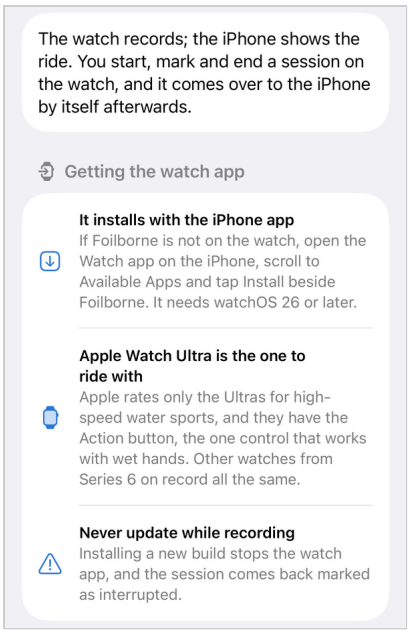
Disciplines (Settings › Disciplines › Modules). **Free discipline** at the top, chosen once; then each module with what it adds. In TestFlight nothing is charged.



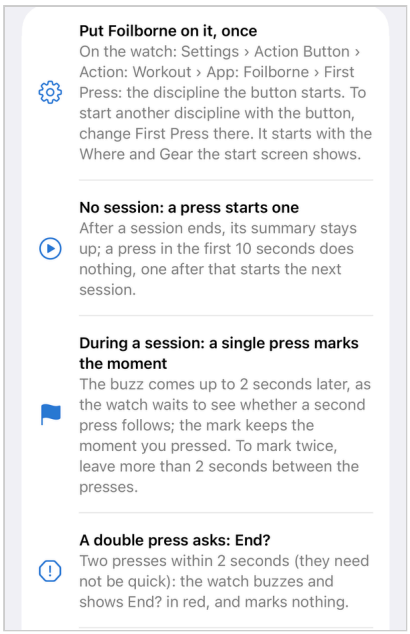
The foot of Disciplines: **All disciplines**, then **Redeem a code**, where the voucher's code goes, and **Restore purchases**.



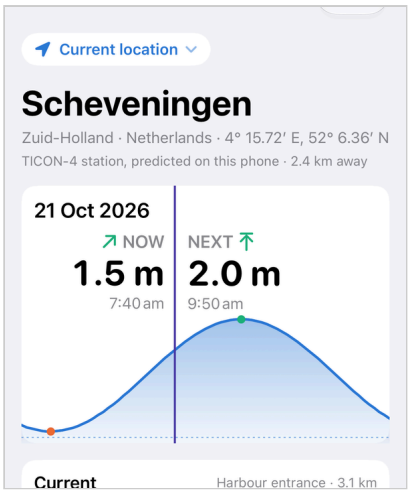
My gear. Add gear for each setup: a name, the all-up weight (you, the board, the foil and what you carry), the motor (none, foil assist, eFoil direct or geared 4:1) and, if you know it, your own take-off speed. The list goes to the watch's start screen.



Using the watch. This guide's watch part, in the app: getting the watch app, what it asks for, starting, the Action button, ending, the pages, the ESC. At the top of Settings, and at the top of the list until your first session is in.



Using the watch, the Action button. The setup in *the watch's Settings app* (Action Button › Workout › Foilborne › First Press), then what a press, a double press and the confirming press do. Keep it open beside the watch while you set the button up.



The tide page (the waves at the top of the list). The nearest gauge, now and next, the curve, the current where a gauge is near, then three days of tide and the highs and lows as a table. **Current location** over the title picks another place. Not for navigation.